Johannes Roth

Machine Learning & Software Engineer

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Available from January 2027

Machine learning and software engineer with experience in applied research, model development, and web applications. Background in taking projects from requirements and experimentation through implementation, deployment, and ongoing maintenance.

EDUCATION

Doctoral Candidate (Dr. rer. nat.)
University of Giessen; May 2022 – Present

M.Sc. Computer Science; grade 1.2
Leipzig University; 2017 – 2021

B.Sc. Business Information Systems; grade 1.5
Leipzig University; 2014 – 2017

TECHNICAL SKILLS

ML & Analysis: Python, SQL, PyTorch, TensorFlow, scikit-learn; computer vision, embeddings, model evaluation, statistical analysis, experimental design.

Engineering: FastAPI, Flask, Django, HTML/CSS, Redis, Docker, Git, CI/CD; data pipelines.

Cloud & Compute: AWS (S3, IAM), Linux, SLURM/HPC.

Languages: German (native), English (fluent).

EXPERIENCE

PhD Researcher May 2022 – Present
Max Planck Institute for Human Cognitive and Brain Sciences & University of Giessen

- Conducted independent research on **vision models and human brain activity**, combining machine learning, statistical analysis, and fMRI experiments.
- Independently developed and maintained **web applications and AWS-backed data services** for sharing research datasets, covering requirements gathering, access control, hosting, and deployment.
- Developed data-processing and model-evaluation pipelines for curating large-scale image datasets and comparing vision models with brain activity.
- Published research at **NeurIPS 2026 (shared first author)** and presented work on experimental design at **CCN 2025 (oral)**.
- Supervised working students and interns.

Research Scientist - Machine Learning in Medicine Jun 2021 – May 2022
ScaDS.AI Dresden/Leipzig

- Developed **ensemble models for MRI brain-tumor segmentation**, with results published in the MICCAI BrainLes workshop proceedings.
- Developed and evaluated mortality-prediction models for clinical data, including uncertainty estimation.

Data Scientist / ML Engineer (Working Student) Nov 2019 – May 2021
CHECK24, Travel Vertical

- Independently designed and deployed a **production computer-vision service that classified hotel photos by season**; built the models and backend and processed a catalogue of **20 million images**.
- Extended the work through prototypes for duplicate detection, image retrieval, and image-quality assessment.
- Introduced incremental processing for recommendation-data updates, reducing update time by approximately **95%**.

Full-Stack Developer (Freelance) Oct 2020 – May 2021
Kimetric UG

- Designed and built **research-group** and **public-dataset websites** using Django.
- Gathered requirements directly from clients and managed Linux hosting, deployment, and ongoing application maintenance.

SELECTED PROJECTS

- **thingsvision** – Contributed to an open-source library for extracting and comparing vision-model features, including memory-efficient extraction, dataset support, and model-integration tests.
- **LAION-fMRI** – Designed the image collection and built research software, the public website, and a download service providing controlled access to a neuroimaging dataset through AWS.
- **re:vision** – Developed the website and data services for a community replication challenge, plus labeling tools used by working students and interns to prepare the dataset.
- **ReLAION-natural** – Published naturalness scores covering **2.1 billion image URLs**, alongside embeddings for a selected natural-image subset, to support dataset selection and visual similarity research.

Find more open-source projects on my GitHub.

SELECTED PUBLICATIONS

- F. P. Mahner*, **J. Roth***, et al. *Characterizing Universal Object Representations Across Vision Models*. **NeurIPS 2026**. Shared first author.
- **J. Roth**, M. N. Hebart. *How to sample the world for understanding the visual system*. **CCN 2025, oral presentation**.
- **J. Roth***, Y. Duan*, et al. *Ten principles for reliable, efficient, and adaptable coding in psychology and cognitive neuroscience*. Communications Psychology, 2025. Shared first author. **CMBB Replication Award (2025)**.
- **J. Roth** et al. *Multi-plane UNet++ Ensemble for Glioblastoma Segmentation*. MICCAI BrainLes 2021 workshop proceedings, published 2022.

EARLIER EXPERIENCE

Data Scientist (Working Student) Oct 2018 – Sep 2019
Webdata Solutions GmbH (now Vistex)

- Developed ML models for matching products across retailers, improving image-based matching accuracy from **12% to 83%** against a classification baseline in an internal evaluation.

Data Analyst (Working Student) Feb 2018 – Aug 2018
Mercateo Services GmbH & Co. KG

- Built data-integration, reporting, and testing workflows for an **AWS-based data warehouse** using Apache NiFi and Pentaho.